

USAID KHAN

Software Engineer - Full-Stack Developer, DevOps & AI-Automation

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SUMMARY

Software Engineer with production experience across Java/Spring Boot backend systems, React/Next.js full-stack applications, and enterprise integration work, supported by hands-on DevOps, cloud, and CI/CD practice. Track record of root-causing data-integrity and concurrency bugs in financial systems, leading multi-screen accessibility and modernization initiatives, and building/validating AI-agent evaluation benchmarks. Comfortable working across the full stack from PostgreSQL/Hibernate/DataJPA query design to JWT authentication to React state management and across the deployment pipeline from Docker/Kubernetes containerization to GitOps-based CI/CD.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), TypeScript, Python, SQL, C

Frontend: React.js, Next.js, Angular (version migration), Redux Toolkit, React Router, react-hook-form, HTML5, CSS3, Tailwind CSS

Cloud & Containers: AWS, Microsoft Azure (incl. Azure Service Bus), Oracle Cloud Infrastructure, Docker, Docker Swarm, Kubernetes, Helm Charts, OpenShift

Testing: JUnit 5, Mockito, Vitest, Playwright (E2E & live-browser audits), Micronaut Test, Integration Testing, pytest

AI / Automation: RAG, Generative AI, Spring AI, LLM Integration, MCP Servers, AI Agents, Coding/CLI Agents, Vector DB, AI-Agent Benchmark Design, Agentic Evaluation

Backend & Frameworks: Spring, Spring Boot, Spring Security, Spring Data JPA, Spring AI, Hibernate (ORM), JDBC, REST APIs, Microservices Architecture, MVC Architecture, JWT, OAuth 2.0, RBAC, JMS/Message-Driven Architecture

Databases: PostgreSQL, MySQL, Oracle, MongoDB, Liquibase (schema migration)

CI/CD & DevOps: GitHub Actions, GitLab CI, Jenkins, GitOps, ElectricFlow, Linux Administration, Bash, Networking (subnetting, NAT, IP allowlisting)

Monitoring & Observability: Prometheus, Grafana, Splunk, Apache Kafka, Audit Logging, IAM, AuthN/AuthZ

Tools & Collaboration: Git, GitHub, GitLab, Maven, Gradle, Jira, ServiceNow, Confluence, Slack, Microsoft Teams, etc..

PROFESSIONAL EXPERIENCE

Software Development Engineer

Data Innovations Technologies Pvt. Ltd.

Apr 2025 – Present

Fintech / Accounting SaaS Platform (Spring Boot & React)

- Root-caused and resolved a Spring transaction-propagation defect in a multi-module accounting platform that caused bank-account creation to report failure while the record was silently committed to PostgreSQL a data-integrity bug in a financial system by realigning transaction boundaries and enforcing explicit rollback.
- Identified and eliminated a recurring falsy-zero pagination defect across 39 independent list/report screens in a React + Redux application after tracing the pattern in a single financial module, restoring correct pagination application-wide.
- Resolved Hibernate LazyInitializationException crashes on production list endpoints by correcting @Transactional scoping, and hardened JWT authentication handling to convert unhandled 500 errors on malformed tokens into correct 401 responses while removing sensitive token values from logs.
- Led a systematic dark/light theme and WCAG-AA accessibility remediation across a 67-screen React frontend, building custom Playwright-based contrast and theme-inversion audit tooling and reducing measured contrast failures from 38 to 0, verified against a 2,374-test Vitest suite and a production build.
- Diagnosed a CSS cascade collision between two design systems (Bootstrap and shadcn/ui) causing incorrect component colors app-wide, and consolidated ~30 screens' duplicated dropdown styling into a single design-token-driven module.
- Added backend (JUnit 5/Mockito) and frontend (Vitest) regression test coverage for 20+ production bug fixes and restored environment-variable-based database credential configuration after it had regressed to hardcoded values.

Healthcare Practice Management CMS & Marketing Platform (Next.js / React)

- Built a database-backed content management system (Postgres, Prisma ORM) for a healthcare clinic's public site and admin dashboard, implementing JWT-based access/refresh-token authentication with hashed refresh tokens and audit logging, appointment booking, and blog/service/team content management.
- Migrated a database-backed, per-request-rendered public site to a fully static, build-time-generated Next.js architecture, eliminating runtime database dependency for all content pages while preserving SEO metadata, sitemap, and structured-data output.
- Diagnosed and fixed an SSR/CSR hydration mismatch in a reduced-motion accessibility hook.
- Resolved a recurring build-and-deploy pipeline failure caused by a native image-processing dependency missing from the serverless output bundle fixed across three iterations (optional-dependency pinning, dynamic import, explicit output-tracing configuration) to stabilize production deployments.
- Engineered a spam-resistant contact form pipeline with schema validation, honeypot fields, per-IP rate limiting, and dual-recipient SMTP delivery.

Enterprise Financial Services Platform (Client Engagement)

- Supported an enterprise retirement/investment-services web application (plan sponsors, compliance, aggregator/data-feed reporting) built on Java/Spring Boot, contributing to core business logic, REST API development, and cross-environment (Dev/QA/ST/AT/Production) validation.
- Investigated and resolved Azure Service Bus message-processing exceptions by diagnosing service-bus logs and implementing corrective JMS/Spring Boot configuration properties; supported containerized workloads on OpenShift and Azure/AWS environments.
- Built integration tests for a microservice using Micronaut Test infrastructure, validating service behavior across multiple environments and investigating test failures.
- Migrated a legacy Angular 13 application to Angular 21 within approximately one week while preserving existing functionality, as part of a broader application-modernization effort.
- Performed production issue investigation, root-cause analysis, and remediation, including application health monitoring, live patching, and coordinated deployment/decommissioning activities; used Splunk for log analysis and Jira/ServiceNow for issue tracking.
- Troubleshooted enterprise network connectivity issues involving subnetting, NAT, and firewall/IP allowlisting for Azure Pipelines and internal services.

Cross-Cutting Practices

- Containerized services with Docker and deployed across Kubernetes and Docker Swarm, with hands-on exposure to AWS, GCP, and Oracle Cloud Infrastructure.
- Automated build, test, and deployment workflows through CI/CD pipelines (Jenkins, GitHub Actions, GitLab, ElectricFlow) using GitOps-based practices, troubleshooting and resolving build/deploy pipeline failures encountered on code pushes to keep releases unblocked.
- Integrated Spring AI/LLM-driven workflows and CLI coding agents into the development process to accelerate delivery.

Software Development Intern

Aug 2024 – Jan 2025

Data Innovations Technologies Pvt. Ltd.

- Contributed to backend applications and API-driven workflows, including REST API integrations, JWT/OAuth 2.0 authentication mechanisms, service-to-service communication, and feature implementation.
- Assisted in containerizing multiple applications with Docker and supported CI/CD build, test, and deployment pipelines in production-like environments.
- Worked extensively in Linux environments for application setup, server configuration, troubleshooting, and deployment management across the full software deployment lifecycle.
- Delivered 10+ enterprise-grade WordPress and e-commerce websites using WordPress, Elementor, WooCommerce, and Shopify across digital platforms.

AI BENCHMARK ENGINEERING(PROJECTS)

- **Handshake AI — Project Dynamo:** Designed and validated a deterministic AI-coding benchmark for forensic recovery of a corrupted content-addressed backup store, covering chunkmap reconstruction, content-defined-chunking parameter inference, snapshot/pack chronology recovery, and historical-forgery detection. Built the Docker/Harbor task environment, verifier, and oracle solution (validated at reward 1.0 with a 0.0 no-op baseline), and debugged GitHub Actions CI and AI-agent evaluation timeout behavior.
- **Snorkel AI — Sentinel Ultra:** Audited and strengthened repository-level software-engineering benchmarks, including C/ARM64 NEON SIMD optimization work on Netflix VMAF (cross-compilation, QEMU execution, Docker, Meson, bit-exact scalar-vs-vectorized validation) and a Janet compiler linting/variable-shadowing task. Identified real SIMD boundary-condition defects, strengthened tests with public-API end-to-end validation, and debugged agent-evaluation and packaging-integrity failures.

CERTIFICATIONS & COURSES

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| ▪ Introduction to Linux (LFS101) - Linux Foundation | ▪ Introduction to GitOps (LFS169) - Linux Foundation |
| ▪ Introduction to Kubernetes (LFS158) - Linux Foundation | ▪ Introduction to DevOps and Site Reliability Engineering (LFS162) - Linux Foundation |
| ▪ Oracle Cloud Infrastructure Foundations Associate (2025) | ▪ SRE Fundamentals: Mastering Site Reliability Engineering |
| ▪ Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | ▪ IBM Cloud Computing Fundamentals |
| ▪ Oracle Cloud Infrastructure AI Foundations | |

EDUCATION

Bachelor of Science in Computer Science BSc(CS) - Grade A+ , CGPA 8.28
Aki's Poona College of Arts,Commerce & Science, Savitribai Phule PuneUniversity

Nov 2022 - Apr 2025

LANGUAGES

English (Fluent) | Hindi (Native) | Marathi (Native)